Tree Frog Anti-Slip Dashboard Pad.mp4

Fixate Gel Pads[1].mp4

Fixate Gel Pads.mp4

Anti-Slip Sticky Mat Renewal.mp4

Sticky Pads.mp4

Tree-frogs rely on the joint efforts of friction and adhesion to stick to a vertical or ceiling-type surface, new videos reveal.

While the creatures rely on their sticky toe pads to hold them up, new footage of the frogs walking upside-down reveal that they increase their grip through friction, by adopting a spread-out stance similar to that of the famous comic-book crime-fighter, Spider-Man.

The method that frogs employ to stick to a ceiling or vertical tree trunk is different to the Velcro-like foot bristles geckos use, the researchers discovered.

Tree frogs use wet adhesion to cling to surfaces – the animals secrete a thin layer of sticky mucus to coat their fleshy toe-pads. According to Jon Barnes at the University of Glasgow in the UK, and colleagues, many believe that this layer of mucus prevents friction from forming between the frogs' feet and a vertical surface – such as a tree trunk – or anything steeper.

- **How are lizards able to climb walls?**
- "The toe pads on the underside of gecko (lizard) feet contain tiny hair-like structures called setae, which stick to surfaces through frictional forces as well as forces between molecules, called Waals. These tiny structures are so strong that the setae (stiff hair) on a single foot can support 20 times the gecko's body weight". Jun 20, 2016

- Source: (https://www.google.co.in/search?dcr=0&source=hp&ei=_3xdWuXpFYH48QWRioCgCA&q=How+house+lezard+climb+on+the+wall%3F&oq=How+house+lezard+climb+on+the+wall%3F&gs_l=psy-ab.3...9240.32833.0.33483.0.0.0.0.0.0.0.0.0..0.0....0...1c.1.64.psy-ab..0.0.0....0.QkM3zQzj8Ss)

- **Introduction:**
- Conventionally engineering- do they have anything to do with art and aesthetics?
- Traditionally engineering education is believed to do with machines and its tools to produce various components for larger products.
- 20th century such concept started changing to give birth to a new direction of *Industrial Design*.
- Design is a branch of applied filed that provided new outlook- *creativity in design*.
- Due to the 21st Century Globalization Policy, the Consumer Products have taken the centre stage of the nation's revenue generation.
- Industrial Design bridges between *art*, *science*, *and technology*. Please remember **Industrial Design is not Engineering Design**. The innovation in design mostly happens while observing other activities from nature. Nature is the greatest source of innovations.

What is ‘Innovation’?

- **Definition of innovation (Webster Dictionary)**

  1. the introduction of something new
  2. a new idea, method, or device **: <u>novelty</u>**

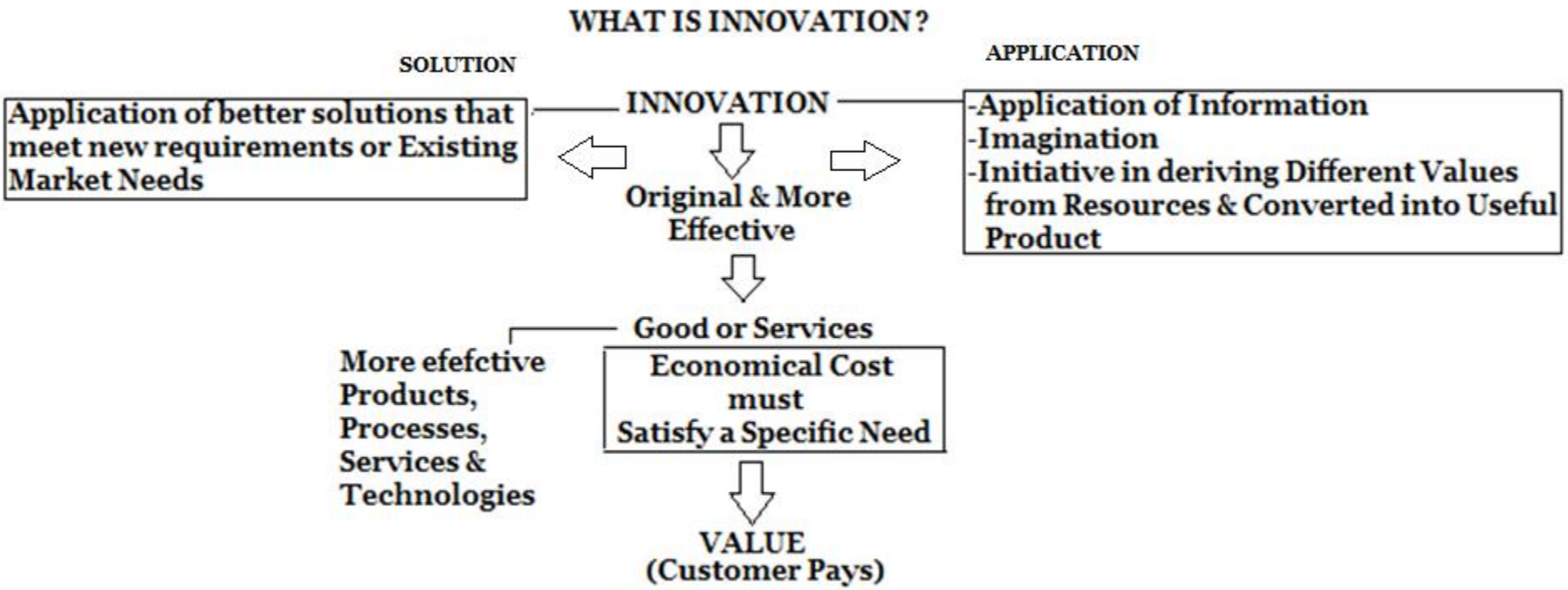
- **“The process of translating an idea or invention into a good or service that <u>creates value</u> or for which customers will pay.”** (Are we doing for the sake of doing?)

<u>The above explanation depends on the following two underlying objectives-</u>

- **<u>Where do we want to be?</u>**
- **<u>What exactly are the ‘preferred situations’ or ‘desired goal’?</u>**
- **<u>How do we get there? What course of action will take us from here to there?</u>**
- **<u>What values it adds to our lives?</u>**

(Prof. John Thackara, <u>‘In the Bubble Designing in A Complex World’</u>, Pub. MIT Press)

INNOVATION



Innovation is more apt to involve the practical implementation of an Invention (i.e new/ improvement)

CCC 510 'CREATIVITY & CONCEPT IN DESIGN'

- **Before trying to understand about anything we need to clarify the following three wards-**
- **Creativity**
- **Concept &**
- **Design**

Male Butterflies Communicate with the help of 'pheromones' between the same species.

Creativity

- Problem Identification could be one of the major components for creative inspiration
- Creativity, Concept and Design, are they related? Creativity is something that is experienced by every individual. If you observe, every leaving being display their ability to display 'creativity'. Never think 'creativity' belongs to who are gifted with some ability. Every individual displays creativity depending on the situation or need. Sometimes you do not recognize your own creative act. It has multifaceted ways of expression.
- Don't you see when insect tries to camouflage in front of prey they apply degree of cognitive intelligence alongwith creative mechanism to hide? Mankind has used creative expression for pleasure, safety, solving problem, comfort, etc.
- Q. Have you seen an animal/ birds/ insects display their creativity?

Creativity in other living beings

Bowerbird

Nest Structure

How to Attract a Bird! - Battle of the Sexes in the Animal World - BBC Earth - BBC.mp4

He Delivers Flowers_ Vogelkop Bowerbird.mp4

- Male Bowerbirds build decorated structures to woo females.
- Tall towers made of sticks resting upon a round mat of dead black moss,
- Decorated with snail shells, acorns, green moss, decorated with fruits, flowers, and severed butterfly wings.
- The males place each item in their bowers with great precision; if the objects are moved, the birds return them to the original arrangement.
- The bird makes calculative choices to decorate the front of the nest. Aesthetic appeal is the most important factor to attract the female partner.
- Hence, not just human being but also other living beings make similar creative choices!